Definition 10.7

A **nonlinear ordinary differential equation** is simply one that is not linear.

If the coefficients of $y, y', y'', \ldots, y^{(n)}$ contain the dependent variable y or its derivatives or if powers of $y, y', y'', \ldots, y^{(n)}$, such as $(y')^2$, appear in the equation, then the differential equation is nonlinear. Also, nonlinear functions of the dependent variable or its derivatives, such as $\sin y$ or $e^{y'}$ cannot appear in a linear equation.

For instance,

(1) $\frac{dy}{dx} = ax^3$, $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ and $\frac{dy}{dx} + p(x)y = q(x)$ are linear differential equations where as $y\frac{dy}{dx} + \sin x = 0$ is a nonlinear differential equation.

(2) $y'' + 2x^3y' = 7xy + x^2$ is a second order linear ODE.

(3) $y'' + y' = \sqrt{x}$ is a second order linear ODE.

(4) $y^2 + y' = \sqrt{x}$ is a first order nonlinear ODE.

(5) $y' = x\sin(y)$ is a first order nonlinear ODE.

(6) $y'' = y\sin(x)$ is a second order linear ODE.

Definition 10.8

If $g(x) = 0$ in (1), then the above equation is said to be **homogeneous**, otherwise it is called **non-homogeneous**.

Remark

If $y_i(x)$, $i = 1, 2$ are any two solutions of homogeneous equation

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x) = 0 \qquad \ldots(2)$$

then $a_n(x)y_i^{(n)}(x) + a_{n-1}(x)y_i^{(n-1)}(x) + \cdots + a_1(x)y_i(x) + a_0(x)y_i(x) = 0$, $i = 1, 2$.

Suppose $u(x) = c_1y_1(x) + c_2y_2(x)$, where c_1 and c_2 are arbitrary constants. Then, it can be easily verified that $u(x)$ is also a solution of (2).

Thus, a first order linear differential equation is written as $y' + p(x)y = f(x)$. A first order differential equation that can't be written like this is nonlinear. Since $y = 0$ is obviously a solution of the homogeneous equation $y' + p(x)y = 0$, we call it the trivial solution. Any other solution is nontrivial. In fact this is true for a general linear homogeneous differential equation as well.

12th_Maths_Vol 2_EM_CH 10_Differential Equation.indd 150 16-12-2022 16:47:12